

Replay-Fidelity: A Structural Audit of Public Conversational AI Datasets for Deterministic Replay

Mitch Baba

DriftForge Systems

February 2026

Public Release — Structural audit only — No behavioral evaluation — No reconstruction — No inference

Abstract

Publicly available conversational datasets are frequently described as multi-turn, session-based, or trajectory-level artifacts. However, many downstream analyses implicitly assume that such artifacts support deterministic, turn-by-turn replay without reconstruction or inferred state. This paper formalizes replay-fidelity as a minimal structural requirement for faithful conversational replay and introduces a deterministic binary audit framework for evaluating structural eligibility.

We apply this framework to over 300,000 records across five widely used public datasets, including HarmBench, HH-RLHF, OASST1, ShareGPT V3, and MultiWOZ 2.1. The audit evaluates explicit turn structure, role attribution, ordering determinism, alternation depth, and continuity completeness without normalization or reconstruction. We find that replay-eligible records are rare under minimal structural criteria and that several artifacts commonly characterized as multi-turn do not structurally support deterministic replay without loss.

These results clarify a measurement boundary between analysis-grade artifacts and those structurally eligible for replay-based trajectory analysis. This study does not evaluate model behavior or alignment quality; rather, it establishes replay-fidelity as a structural precondition for deterministic conversational analysis.

1 Introduction

Public conversational artifacts are frequently described as multi-turn. However, the structural conditions required for deterministic replay are rarely examined explicitly.

Replay-based trajectory analysis presupposes that conversational records can be replayed without reconstruction, inference, or cross-record stitching. Absent explicit structural guarantees, conversational artifacts may not support deterministic turn-by-turn evaluation.

This work formalizes replay-fidelity as a binary structural predicate over individual records and applies it to publicly released conversational datasets to measure replay eligibility.

This study evaluates structural properties only. It does not assess model behavior, alignment performance, safety outcomes, or conversational utility.

Governance and observation applicability. Replay-fidelity is directly applicable to AI governance, safety auditing, red-team evaluation, and post-deployment behavioral observation systems that rely on deterministic replay of conversational trajectories. Where structural

replay is undefined, replay-only governance or observational methodologies cannot be applied without reconstruction or inference. Replay-fidelity therefore formalizes a structural precondition for replay-based oversight and evaluation systems.

2 Related Work

A substantial body of research has introduced and analyzed conversational datasets for dialogue modeling, evaluation, and safety benchmarking. Datasets such as MultiWOZ 2.1 have been widely used for task-oriented multi-turn dialogue research. Other artifacts, including HH-RLHF, OASST1, HarmBench, and ShareGPT derivatives, are frequently referenced in discussions of conversational alignment, red-team evaluation, and large-scale model behavior.

Prior work has primarily focused on semantic performance, policy adherence, or alignment outcomes. Multi-turn evaluations typically assume that conversational trajectories can be reconstructed or analyzed at the interaction level, often without formalizing the structural preconditions required for deterministic replay.

This work differs in scope. Rather than evaluating model behavior, alignment quality, or attack success, we examine the structural properties of publicly available artifacts themselves. Specifically, we formalize replay-fidelity as a minimal structural requirement for deterministic turn-by-turn analysis and empirically assess whether widely used datasets satisfy those requirements without reconstruction.

3 Contributions

This paper makes the following contributions:

1. Formalization of replay-fidelity as a structural predicate for deterministic conversational replay.
2. Introduction of a deterministic binary structural audit framework.
3. Empirical audit of over 300,000 records across five widely used public conversational datasets.
4. Demonstration that replay-eligible records are rare under minimal structural criteria.

5. Clarification of a measurement boundary between analysis-grade and replay-grade conversational artifacts.

4 Formal Definition of Replay-Fidelity

Replay-fidelity is defined as a binary predicate over individual records.

A record is **REPLAY_ELIGIBLE** if and only if all structural invariants are satisfied. Otherwise, the record is **REPLAY_INELIGIBLE**.

The predicate evaluates record structure only and does not interpret conversational content.

4.1 Structural Invariants

A record satisfies replay-fidelity if all of the following conditions hold:

- **Explicit Turn Enumerability.** Turns are explicitly enumerable as discrete containers.
- **Explicit Role Attribution.** Each turn contains an explicit role field or a deterministically declared role mapping.
- **Deterministic Ordering.** Turn order is unambiguous and reproducible.
- **Alternation Depth.** The record contains at least four alternating turns: user $\rightarrow$ assistant $\rightarrow$ user $\rightarrow$ assistant.
- **Record Boundary Integrity.** No cross-record stitching or inference is required to establish conversational continuity.

Replay-fidelity is binary. No graded or partial classification is permitted under the predicate.

Alternative alternation thresholds define distinct predicates and are outside the scope of this study.

5 Audit Methodology

The replay-fidelity audit was conducted using a deterministic Python implementation relying exclusively on the Python standard library.

Each record is evaluated independently. The implementation performs invariant checks without reconstruction, inference, heuristic parsing, or cross-record aggregation.

5.1 Binary Predicate and Diagnostic Categories

The implementation emits three diagnostic categories:

- `REPLAY_OK`
- `REPLAY_PARTIAL`
- `REPLAY_INVALID`

These categories reflect structural diagnostic outcomes.

Under the formal replay-fidelity predicate:

- `REPLAY_OK` corresponds to **`REPLAY_ELIGIBLE`**.
- Both `REPLAY_PARTIAL` and `REPLAY_INVALID` correspond to **`REPLAY_INELIGIBLE`**.

`REPLAY_PARTIAL` indicates that some structural conditions are satisfied but at least one invariant is violated. Under the binary predicate, any invariant violation yields structural ineligibility.

6 Datasets Evaluated

The audit was applied to publicly released conversational artifacts:

- OpenAssistant OASST1 (train and validation splits)
- Anthropic HH-RLHF (train and test splits)
- ShareGPT V3 (cleaned split)
- HarmBench (raw and canonicalized)
- MultiWOZ 2.1

All artifacts were evaluated as released. No structural normalization or schema modification was performed beyond deterministic canonicalization where explicitly documented.

7 Empirical Results

Table 1 — Replay Eligibility by Dataset

Dataset	Mode	Records	Eligible	Ineligible	% Eligible
HarmBench	Explicit	537	0	537	0.00%
HH-RLHF (Train)	Explicit	160,800	0	160,800	0.00%
HH-RLHF (Test)	Explicit	8,552	0	8,552	0.00%
MultiWOZ 2.1	Declared	10,438	10,296	142	98.64%
MultiWOZ 2.1	Explicit	10,438	0	10,438	0.00%
ShareGPT V3	Explicit	94,145	69,245	24,900	73.55%
OASST1 Train	Explicit	84,437	0	84,437	0.00%
OASST1 Validation	Explicit	4,401	0	4,401	0.00%

Eligibility rates differed substantially across artifacts and were strongly correlated with schema structure. Across evaluated artifacts, structural failure modes cluster by invariant type.

Table 2 — Failure Mode Distribution

Dataset	NO_ENUM_TURNS	ALT_FAIL	ROLE_FAIL	TRUNC	EMPTY
HarmBench	0	537	0	0	0
HH-RLHF (Train)	160,800	0	0	0	0
HH-RLHF (Test)	8,552	0	0	0	0
MultiWOZ (Declared)	0	4	0	138	0
MultiWOZ (Explicit)	0	0	10,438	0	0
ShareGPT	0	23,593	757	0	550
OASST1 Train	84,437	0	0	0	0
OASST1 Validation	4,401	0	0	0	0

Column key: NO_ENUM_TURNS = NO_ENUMERABLE_TURNS; ALT_FAIL = INSUFFICIENT_ALTERNATION; ROLE_FAIL = MISSING_REQUIRED_ROLE; TRUNC = TRUNCATION; EMPTY = EMPTY_TURN_LIST.

Role Attribution Mode Sensitivity (MultiWOZ Case)

MultiWOZ 2.1 exhibits a marked contrast between declared role-mapping mode and explicit-only evaluation. Under declared mode, role attribution is deterministically derived from documented schema conventions, resulting in high replay eligibility (98.64%). Under explicit-only evaluation—where role fields must be present directly within turn containers—all records fail the explicit role attribution invariant.

This contrast does not reflect inconsistency in the dataset itself. Rather, it illustrates that replay-fidelity depends on whether role attribution is encoded structurally within record containers or derived from documented schema conventions. In declared mode, the role-mapping rule is frozen and applied uniformly, preserving determinism in classification. In explicit-only mode, the absence of per-turn role fields constitutes a structural invariant violation.

The MultiWOZ case demonstrates that replay eligibility can be contingent on schema encoding decisions even when underlying conversational content remains unchanged. Replay-fidelity therefore evaluates structural representation, not conversational substance.

8 Structural Archetype Analysis

The audit reveals recurring structural archetypes:

- **Archetype I — Flattened Transcript Structures.** No explicit turn containers. Replay structurally undefined.
- **Archetype II — Explicit Turns, Insufficient Alternation.** Turn containers present but fail minimum alternation depth.
- **Archetype III — Fully Structured Linear Dialogue Containers.** Explicit turns, roles, ordering, and sufficient alternation.
- **Archetype IV — Role-Encoding Sensitivity.** Replay eligibility contingent on schema-level role encoding.

Replay eligibility correlates with record representation rather than conversational content.

9 Structural Preconditions for Deterministic Replay

Replay-based trajectory analysis presupposes:

$$\text{Replay}(r) = 1$$

If:

$$\text{Replay}(r) = 0$$

the conversational trajectory is structurally undefined under replay-only constraints.

Replay-fidelity therefore defines a structural precondition for deterministic replay-only conversational evaluation.

It does not evaluate semantic quality, behavioral safety, alignment performance, or conversational richness.

10 Structural Implications for Conversational Dataset Design

Replay eligibility correlates with:

- Explicit turn containerization
- Deterministic role attribution encoding
- Sufficient alternation depth
- Record-level conversational encapsulation

These observations are structural and descriptive. Replay-fidelity evaluates only whether deterministic replay is structurally defined.

11 Conclusion

Replay-fidelity formalizes a binary structural predicate for deterministic conversational replay.

Empirical evaluation demonstrates that replay eligibility varies substantially across publicly released artifacts and is strongly determined by schema design.

Replay-fidelity isolates structural replay capability from conversational semantics and provides a reproducible framework for evaluating replay eligibility.

12 Limitations

Replay-fidelity evaluates structural properties only. It does not assess:

- Model behavior
- Alignment quality
- Safety outcomes
- Conversational effectiveness
- Dataset utility for other research purposes

The four-turn threshold is defined as minimally sufficient for deterministic replay-based trajectory observation under this predicate.

This study evaluates publicly released artifacts only. It does not evaluate internal logging systems, proprietary evaluation pipelines, production monitoring infrastructure, or private dataset releases. No claims are made regarding the structural properties of non-public conversational artifacts.

This study evaluates structural properties of publicly available conversational artifacts. It does not evaluate model behavior, alignment quality, policy compliance, or attack success. Structural replay eligibility is treated strictly as a measurement precondition, not as a proxy for model safety or system robustness.

The audit operates on released dataset artifacts as distributed. It does not reconstruct missing turns, infer implicit structure, normalize formats, or attempt semantic repair. Datasets that may support replay internally within their originating organizations may fail structural eligibility when assessed using only publicly available artifacts.

This work does not claim that replay-fidelity is required for all forms of conversational evaluation. Many analyses—such as single-turn classification, pairwise comparison, or prompt-

response benchmarking—remain valid without full replay structure. The results apply only to deterministic turn-by-turn replay analysis without reconstruction.

13 Reproducibility

The replay-fidelity audit was conducted using a deterministic Python implementation (`replay_fidelity_audit.py`) relying exclusively on the Python standard library.

All evaluated artifacts were stored under controlled directory paths. For each dataset file evaluated, a SHA256 hash was computed and recorded in the acquisition log. These hashes uniquely identify the exact artifact versions used in this study and ensure chain-of-custody reproducibility.

Replay-fidelity evaluation is deterministic given:

- A fixed dataset release
- A fixed role-mapping mode
- A frozen invariant definition

Reproduction of results requires:

- The original dataset artifacts matching the recorded SHA256 hashes
- The released audit script
- The invariant definition specified in this paper

No stochastic components, heuristic reconstruction, or cross-record aggregation are used.

A Appendix A — Formal Predicate

Let R denote the set of all conversational records.

Let $E(r)$ be a deterministic extraction function mapping a record $r \in R$ to an ordered sequence of turns:

$$E(r) = \langle t_1, t_2, \dots, t_k \rangle$$

where each turn t_i has (at minimum) a role label $\rho(t_i) \in \{user, assistant\}$ and a content payload $c(t_i)$. System turns may exist but do not satisfy the alternation requirement.

Define invariant checks as boolean-valued functions:

$$I_1(r) = \text{ExplicitTurnEnumerability}(r)$$

$$I_2(r) = \text{ExplicitRoleAttribution}(r)$$

$$I_3(r) = \text{DeterministicOrdering}(r)$$

$$I_4(r) = \text{AlternationDepth}(r)$$

$$I_5(r) = \text{RecordBoundaryIntegrity}(r)$$

Define the failure function:

$$F(r) = \neg I_1(r) \vee \neg I_2(r) \vee \neg I_3(r) \vee \neg I_4(r) \vee \neg I_5(r)$$

The replay predicate is defined as:

$$\text{Replay}(r) = \begin{cases} 1 & \text{if } F(r) = 0 \\ 0 & \text{otherwise} \end{cases}$$

A record is replay-eligible if and only if $\text{Replay}(r) = 1$. Under the binary predicate, any invariant violation implies replay-ineligibility.

B Appendix B — Failure Mode Taxonomy

Diagnostic codes:

- NO_ENUMERABLE_TURNS
- EMPTY_TURN_LIST
- MISSING_REQUIRED_ROLE
- INSUFFICIENT_ALTERNATION
- TRUNCATION_OR_SUMMARY_DETECTED

These codes describe structural conditions only.

Appendix C: Dataset Hashes (SHA256)

Anthropic HH-RLHF (HuggingFace Export)

Train split (train.jsonl)

SHA256: 65208ab6f9a24ca1dab22e4444d09da22f052b6106c122d8b80432c261e486ad

Test split (test.jsonl)

SHA256: 80c2d33f5e9c18305f03e06e36a28dba75fecda48c2ecb240d2d4c3a9e1dd5f6

OpenAssistant OASST1

Train split (train.jsonl)

SHA256: d6b0bd645e51b7241a2d2fd8ea6a7d0d26aeeab08aabe871d2ca9d151c7a1ed1

Validation split (validation.jsonl)

SHA256: d0c97a45cac81bb77ee952cab5a19cae41c9bbda19d430a68a1637c25a3b905d

ShareGPT V3

File: ShareGPT_V3_unfiltered_cleaned_split.json

SHA256: 35f0e213ce091ed9b9af2a1f0755e9d39f9ccec34ab281cd4ca60d70f6479ba4

HarmBench

Raw CSV (harmbench_behaviors.csv)

SHA256: 97e0dc879b66c7b664deb2a9c2e5bd78b647b84e16a9a43ba95d0117e3b35e46

Canonical JSONL (harmbench_canonical.jsonl)

SHA256: 1c1687aace2fa3143f85b737a6fcb4da1251b0f04d04c4a22c908cfef7550d48

MultiWOZ 2.1

File: MultiWOZ-data.json

SHA256: 8be37ba1cb5b5a35943f32d4dbe03c5017dd88e15716f74987f60e0ece37851c